

RANK-DEPTH ADAPTIVE BLOCK MATRIX PRODUCTS FOR QUANTUM OPERATOR LEARNING

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Abstract. Learning a parameterized quantum operator from many correlated preparations requires block products of the form $B^\top f(A(\theta))B$ and their parameter derivatives. Treating the columns independently discards correlation, whereas using the complete block can spend Hamiltonian applications on numerically redundant preparations. We introduce a preflight rank-depth rule that selects a truncated singular-value factorization of B and a block-Krylov depth before the first Hamiltonian application. The rule combines three deterministic ingredients: a uniform spectral enclosure valid on a prescribed parameter box, an explicit bound for the effect of probe compression on every loss-gradient component, and positive Taylor majorants for the remaining matrix-function and Fréchet-derivative tails. In exact arithmetic, the resulting Euclidean bound certifies the full gradient error; the implementation separately records loss of orthogonality. Tests on four checksum-authenticated PARSEC density-functional-theory Hamiltonians, of dimensions 769–19,896, select ranks five or six from an eight-column block and depth 24 at tolerance 10^{-3} . This reduces the Hamiltonian-vector equivalent count by 25–37.5 percent relative to an uncompressed block at the same depth and by factors of 8–9.6 relative to coordinatewise finite differences. Paired CPU timings favor the proposed method on three matrices by factors 1.36–2.04; on the largest matrix the uncompressed block is 1 percent faster. The calculation is a numerical operator benchmark: the Hamiltonians are authentic, while the parameter directions, preparations, and targets are reproducible constructed quantities.

1. Introduction. Matrix functions provide a compact description of quantum propagation, thermal response, filtering, and spectral observables [1–3]. A recurring inverse problem observes several small matrices

$$F_\ell(\theta) = B^\top f_\ell(A(\theta))B, \quad \ell = 1, \dots, s, \quad (1.1)$$

where $A(\theta)$ is too large to factor or diagonalize, $B \in \mathbb{R}^{n \times b}$ contains preparations or probes, and $b \ll n$. The loss depends on every entry of every block product, so the derivative is not a collection of unrelated scalar derivatives.

Krylov projection replaces the large matrix function by a function of a small Hessenberg or tridiagonal matrix [2, 4, 5]. Block Krylov methods reuse one projected space for several right-hand sides and have matrix-function error analyses tailored to that setting [6, 7]. Fréchet derivatives can likewise be approximated through block or tensorized Krylov spaces [8–10]. More recently, Singh, Barros, and Li derived a stable forward-only gradient approximation for scalar bilinear forms of Hermitian matrix functions [11]. Their result addresses the differentiation bottleneck without backpropagating through Lanczos.

The present problem has a distinct resource question. Correlated columns of B may have numerical rank $r < b$. Reducing the block width lowers the cost of every subsequent sparse matrix–matrix product, but it also perturbs the observed block and its gradient. Moreover, checking successive Krylov depths after each extension can consume more time in reduced eigendecompositions and direction projections than it saves in Hamiltonian applications. Our initial implementation exhibited precisely this behavior. A useful algorithm must therefore decide both rank and depth without repeatedly executing the large operator.

We develop such a rule for the cosine response $f_\ell(x) = \cos(t_\ell x)$ and real-symmetric affine families. Its contributions are as follows.

1. A row-sum construction gives a uniform, noniterative enclosure $\|A(\theta)\|_2 \leq \rho < 1$ over a complete parameter box. It avoids using an unconverged extremal Ritz value as a certificate.
2. A probe-compression theorem bounds both the block response and its Fréchet derivative after replacing B by a truncated SVD B_r .
3. Polynomial exactness converts positive cosine tails into a gradient certificate depending only on registered inputs. A conservative residual upper bound makes the certificate available before any Hamiltonian product.

4. A preflight algorithm chooses the least SVD rank within a compression budget and the first Krylov depth satisfying the full gradient tolerance. It then executes one compressed block factorization.
5. An authenticated CPU study reports deterministic work, observed error, certificate slack, all timing repetitions, and a negative timing result on the largest matrix.

The method is classical numerical linear algebra for a quantum inverse problem. It does not claim a quantum computational advantage or infer chemical parameters from experimental measurements.

2. Block operator-learning objective. Let

$$A(\theta) = H_0 + \sum_{j=1}^p \theta_j D_j, \quad A(\theta) = A(\theta)^\top, \quad (2.1)$$

and let the registered parameter domain be $\Theta = \{\theta : |\theta_j| \leq a_j\}$. For observed symmetric targets $Y_\ell \in \mathbb{R}^{b \times b}$, consider

$$\Phi(\theta) = \frac{1}{2s} \sum_{\ell=1}^s \|F_\ell(\theta) - Y_\ell\|_F^2, \quad (2.2)$$

with F_ℓ defined in Eq. (1.1). The j th gradient component is

$$g_j(\theta) = \frac{1}{s} \sum_{\ell=1}^s \langle F_\ell(\theta) - Y_\ell, B^\top \mathcal{L}_{f_\ell}(A(\theta), D_j) B \rangle_F, \quad (2.3)$$

where $\mathcal{L}_f(A, D)$ denotes the Fréchet derivative of f at A in direction D [1, 12]. Equation (2.3) contains sb^2 response entries but only one large operator family.

2.1. Uniform parameter-box scaling. The proof requires a spectral interval known before the Krylov calculation. The following construction uses quantities that are exact for the stored sparse matrix and diagonal directions.

Lemma 1 (Uniform spectral enclosure). *Suppose $H_0 = H_0^\top$, $D_j = D_j^\top$, and*

$$\|H_0\|_\infty + \sum_{j=1}^p a_j \|D_j\|_2 \leq \rho. \quad (2.4)$$

Then $\|A(\theta)\|_2 \leq \rho$ for every $\theta \in \Theta$. If each D_j is diagonal, its norm in Eq. (2.4) is the largest absolute diagonal entry.

Proof. Symmetry gives $\|H_0\|_2 \leq \sqrt{\|H_0\|_1 \|H_0\|_\infty} = \|H_0\|_\infty$. Hence, for $\theta \in \Theta$,

$$\|A(\theta)\|_2 \leq \|H_0\|_2 + \sum_{j=1}^p |\theta_j| \|D_j\|_2 \quad (2.5)$$

$$\leq \|H_0\|_\infty + \sum_{j=1}^p a_j \|D_j\|_2 \leq \rho. \quad (2.6)$$

For diagonal D_j , the spectral norm equals the maximum absolute diagonal entry. $\square$

Our benchmark scales the authenticated Hamiltonian so that $\|H_0\|_\infty = 0.8$, uses $p = 3$, $a_j = 0.5$, and constructs diagonal directions with norm at most 0.12. Thus $\rho \leq 0.98$ without an eigensolver.

3. Rank–depth preflight method.

3.1. Probe-block compression. Let the thin SVD be

$$B = U\Sigma W^\top, \quad B_r = U_r \Sigma_r W_r^\top, \quad (3.1)$$

and write $\delta_r = \|B - B_r\|_F$. The implementation forms a Krylov basis from $C_r = U_r \Sigma_r$ and lifts each $r \times r$ projected product back to $b \times b$ with W_r . Consequently, the output retains its original shape.

Theorem 1 (Probe-compression perturbation). *Let $A = A^\top$, $D = D^\top$, and $\text{spec}(A) \subset [-\rho, \rho]$. Define*

$$\alpha_r = 2 \|B\|_2 \delta_r + \delta_r^2. \quad (3.2)$$

For $f_t(x) = \cos(tx)$,

$$\|B^\top f_t(A)B - B_r^\top f_t(A)B_r\|_F \leq \alpha_r, \quad (3.3)$$

$$\|B^\top \mathcal{L}_{f_t}(A, D)B - B_r^\top \mathcal{L}_{f_t}(A, D)B_r\|_F \leq |t| \|D\|_2 \alpha_r. \quad (3.4)$$

Proof. Put $E = B - B_r$. Because $\|f_t(A)\|_2 \leq 1$,

$$\begin{aligned} & B^\top f_t(A)B - B_r^\top f_t(A)B_r \\ &= E^\top f_t(A)B_r + B_r^\top f_t(A)E + E^\top f_t(A)E. \end{aligned} \quad (3.5)$$

The Frobenius–spectral product inequalities give $2 \|B_r\|_2 \|E\|_F + \|E\|_F^2 \leq \alpha_r$. For a symmetric argument, differentiate $e^{it(A+sD)}$ at $s = 0$ by the Duhamel formula. Unitary invariance gives $\|\mathcal{L}_{e^{it(\cdot)}}(A, D)\|_2 \leq |t| \|D\|_2$; averaging the $+t$ and $-t$ exponentials therefore gives

$$\|\mathcal{L}_{f_t}(A, D)\|_2 \leq |t| \|D\|_2. \quad (3.6)$$

Repeating the expansion with $f_t(A)$ replaced by $\mathcal{L}_{f_t}(A, D)$ proves Eq. (3.4). $\square$

Let $c_B = \|B\|_2 \|B\|_F$ and $c_r = \|B_r\|_2 \|B_r\|_F$. Applying Theorem 1 to Eq. (2.3) gives the preflight compression component

$$\chi_{j,r} = \frac{\alpha_r \|D_j\|_2}{s} \sum_{\ell=1}^s |t_\ell| (c_B + c_r + \|Y_\ell\|_F). \quad (3.7)$$

This quantity requires only the SVD of B , target norms, and direction norms.

3.2. Block Krylov projection. For $m \geq 2$, let

$$\mathcal{K}_m(A, C_r) = \text{span}\{C_r, AC_r, \dots, A^{m-1}C_r\}, \quad (3.8)$$

and let V_m be an orthonormal basis, with $T_m = V_m^\top A V_m$ and $\hat{C} = V_m^\top C_r$. The projected block response is

$$\hat{F}_{\ell,m,r} = W_r \hat{C}^\top \cos(t_\ell T_m) \hat{C} W_r^\top. \quad (3.9)$$

For direction D_j , the derivative uses $\hat{D}_j = V_m^\top D_j V_m$ and the corresponding reduced Fréchet derivative. No $n \times n$ derivative is formed.

Theorem 2 (Block polynomial exactness). *For every polynomial q of degree at most m ,*

$$C_r^\top q(A)C_r = \hat{C}^\top q(T_m)\hat{C}, \quad (3.10)$$

$$C_r^\top \mathcal{L}_q(A, D)C_r = \hat{C}^\top \mathcal{L}_q(T_m, V_m^\top D V_m)\hat{C}. \quad (3.11)$$

Proof. For $0 \leq k \leq m-1$, $A^k C_r = V_m T_m^k \hat{C}$. If $0 \leq k \leq m$, choose $u + v = k$ with $u, v \leq m-1$. Symmetry yields

$$C_r^\top A^k C_r = (A^u C_r)^\top A^v C_r = \hat{C}^\top T_m^k \hat{C}. \quad (3.12)$$

For a monomial $q(x) = x^k$,

$$\mathcal{L}_q(A, D) = \sum_{u=0}^{k-1} A^u D A^{k-1-u}. \quad (3.13)$$

When $k \leq m$, both powers in every summand are at most $m-1$; substituting $A^u C_r = V_m T_m^u \hat{C}$ on both sides of D proves Eq. (3.11). Linearity completes both statements. $\square$

3.3. A preflight gradient certificate. For degree m and radius ρ , define the positive tails

$$R_m^{(0)}(t, \rho) = \sum_{\substack{\nu > m \\ \nu \text{ even}}} \frac{|t|^\nu \rho^\nu}{\nu!}, \quad (3.14)$$

$$R_m^{(1)}(t, \rho) = \sum_{\substack{\nu > m \\ \nu \text{ even}}} \frac{\nu |t|^\nu \rho^{\nu-1}}{\nu!}. \quad (3.15)$$

The code bounds each series by its first term divided by one minus the largest subsequent term ratio and converts the high-precision result to binary64 with upward rounding.

Theorem 3 (Rank–depth preflight certificate). *Assume Lemma 1. Let $\hat{g}_{m,r}$ be the gradient obtained from B_r and the projected products in Eq. (3.9). Define*

$$\begin{aligned} \kappa_{j,m,r} = \frac{1}{s} \sum_{\ell=1}^s [2c_r R_m^{(0)}(t_\ell, \rho) c_r |t_\ell| \|D_j\|_2 \\ + (c_r + \|Y_\ell\|_F) 2c_r \|D_j\|_2 R_m^{(1)}(t_\ell, \rho)], \end{aligned} \quad (3.16)$$

and $\eta_{j,m,r} = \chi_{j,r} + \kappa_{j,m,r}$. Then

$$\|g - \hat{g}_{m,r}\|_2 \leq \left(\sum_{j=1}^p \eta_{j,m,r}^2 \right)^{1/2}. \quad (3.17)$$

Every quantity on the right is available before applying $A(\theta)$.

Proof. Let c_m be the degree- m Taylor polynomial of f_t . Theorem 2 cancels its full and projected contributions. The remaining tails and $\|T_m\|_2 \leq \|A\|_2 \leq \rho$ give

$$\left\| B_r^\top f_t(A) B_r - \hat{F}_{t,m,r} \right\|_F \leq 2c_r R_m^{(0)}(t, \rho), \quad (3.18)$$

$$\left\| B_r^\top \mathcal{L}_{f_t}(A, D_j) B_r - \hat{J}_{j,t,m,r} \right\|_F \leq 2c_r \|D_j\|_2 R_m^{(1)}(t, \rho). \quad (3.19)$$

Furthermore, $\|B_r^\top \mathcal{L}_{f_t}(A, D_j) B_r\|_F \leq c_r |t| \|D_j\|_2$, and $\|\hat{F}_{t,m,r} - Y\|_F \leq c_r + \|Y\|_F$. Insert and subtract the compressed exact gradient in Eq. (2.3). Theorem 1 and Cauchy–Schwarz give $\chi_{j,r}$ for the first difference. Equations (3.18)–(3.19) give $\kappa_{j,m,r}$ for the second. The triangle inequality gives the component bound, and summing its squares proves Eq. (3.17). $\square$

Corollary 1 (First-feasible resource rule). *Fix $0 < \gamma < 1$, tolerance ε , and increasing depth grid $\mathcal{M}$. Choose the smallest r satisfying $\|\chi_r\|_2 \leq \gamma\varepsilon$, then the first $m \in \mathcal{M}$ satisfying $\|\eta_{m,r}\|_2 \leq \varepsilon$. The gradient returned by one rank- r , depth- m factorization satisfies $\|g - \hat{g}_{m,r}\|_2 \leq \varepsilon$ in exact arithmetic.*

The implementation uses $\gamma = 0.2$. Finite precision is audited by $\|V_m^\top V_m - I\|_2$; the theorem is not presented as an interval arithmetic result for the computed basis.

Preflight rank–depth block product. (1) Authenticate the sparse matrix and construct a parameter box satisfying Lemma 1. (2) Compute one thin SVD of B and choose the first rank r whose compression-gradient bound uses at most $\gamma\varepsilon$. (3) Evaluate Eq. (3.17) over the registered depth grid and select the first feasible m . These three steps require no Hamiltonian application. (4) Build one fully reorthogonalized basis V_m from $C_r = U_r \Sigma_r$. (5) Project A and every D_j , evaluate the reduced cosine and Loewner Fréchet matrices, lift with W_r , and return the loss, gradient, certificate, action ledger, and orthogonality audit.

3.4. Computational contract. Let $\mathcal{C}_A(n, r)$ denote the cost of one matrix-free application of A to r columns. The implemented fully reorthogonalized calculation costs

$$O(m\mathcal{C}_A(n, r) + nm^2r^2 + pnm^2r^2 + sp(mr)^3), \quad (3.20)$$

Table 1: Authenticated operator inputs. The displayed radius is the uniform parameter-box bound, not an estimated extremal eigenvalue.

matrix	n	nnz	role	ρ
Si2	769	17,801	development	0.979996
SiH4	5,041	171,903	confirmation	0.980000
benzene	8,219	242,669	confirmation	0.980000
Si5H12	19,896	738,598	confirmation	0.980000

where the terms respectively represent operator products, reorthogonalization, diagonal-direction projection, and the present reduced Fréchet implementation. The dominant basis memory is $O(nmr)$; no dense $n \times n$ matrix is formed. The deterministic Hamiltonian-vector equivalent count is mr , compared with mb for an uncompressed block at the same depth and $2pmb$ for coordinatewise central differences. Equation (3.20) makes no claim that a smaller r must reduce wall time on every sparse kernel.

4. Authenticated numerical study.

4.1. Data and protocol. The matrices are real-space density-functional-theory Hamiltonians from the PARSEC group of the SuiteSparse Matrix Collection [13, 14]. Each compressed archive is downloaded from the collection’s HTTPS endpoint and checked against a pinned SHA-256 digest before Matrix Market parsing. Table 1 separates the development matrix from three confirmation matrices.

The parameter directions are three Gaussian diagonal fields with norm at most 0.12. The eight probe columns are overlapping Gaussian index envelopes centered near the largest diagonal magnitude of the rescaled operator. Neither construction uses atomic coordinates, and neither is labeled experimental. Targets are the depth-28 full-block products at $\theta_\star = (0.32, -0.24, 0.16)$ for times (2, 4, 6). Gradients are evaluated at $\theta_0 = (0.06, -0.03, 0.02)$. Changing the target and gradient references from depth 28 to depth 26 changes them by at most 1.7×10^{-14} and 3.1×10^{-15} , respectively, over the four matrices.

The registered tolerance is 10^{-3} and the depth grid is $\{4, 6, \dots, 28\}$. Comparators are the uncompressed eight-column block at the selected depth, a repeated-candidate rule that recomputes reduced derivatives at every depth, and coordinatewise central differences of losses recomputed from independent full-block factorizations. Timings use Python 3.13.12, NumPy 2.4.3, one Apple M4 Pro CPU, and 24 GB memory. The proposed and same-depth full-block methods use nine paired repetitions with alternating execution order. Data loading is excluded; SVD, selection, factorization, projected directions, and gradient assembly are included.

4.2. Probe rank and depth adaptation. Figure 1 shows that the constructed block is strongly correlated but not exactly low rank. The 10^{-3} protocol selects rank five for SiH4 and rank six for the other matrices. The discarded singular tail is therefore explicitly charged to Eq. (3.7); it is not treated as zero.

The rank–depth map in Fig. 2 is the central adaptation diagnostic. For maximum time two, selected depths range from 10 to 12; at maximum time eight they range from 28 to 32. Tightening tolerance changes the selected rank from five to six. Thus the rule does not merely reproduce one hard-coded depth.

4.3. Certificate and observed accuracy. Figure 3 exposes rather than suppresses certificate slack. Observed errors decrease below 10^{-10} by depth 8–10 and approach the reference floor by depth 14, whereas the preflight rule selects depth 24. The certificate is therefore a uniform safety contract, not an estimator of the realized error. Its purpose is to remove reference-dependent stopping from the algorithm.

All errors in Table 2 lie below their bounds. The selected work is 120 for SiH4 and 144 for the other matrices, compared with 192 for the uncompressed same-depth block. The exact reduction is therefore 37.5 percent and 25 percent, respectively. Basis storage is nmr floating-point entries; for Si5H12 this is 2,865,024 entries, compared with 395,850,816 entries in a dense square matrix.

4.4. Runtime and a negative case. The proposed implementation is 1.36, 1.65, and 2.04 times faster than the same-depth full block on Si2, SiH4, and benzene. It is 0.99 times as fast on Si5H12: the deterministic work reduction does not overcome the host’s sparse block-kernel and dense-projection

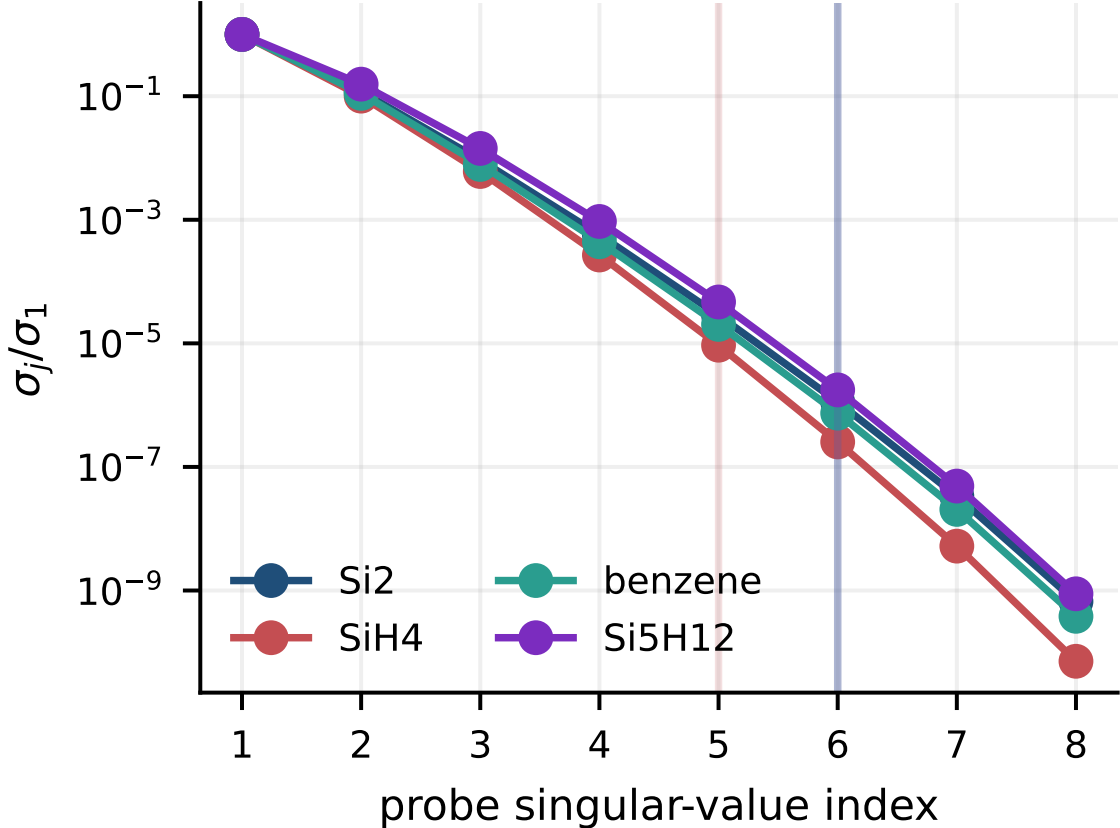


Fig. 1: Normalized singular values of each eight-column numerical probe block. Vertical lines mark the selected rank. The decay, rather than matrix dimension, controls the compression part of the preflight certificate.

behavior in that case. Central differences are 3.88–10.67 times slower and use 1,152 Hamiltonian-vector equivalents, versus 120–144 for preflight. Their gradient errors are 1.18×10^{-11} – 2.70×10^{-11} , compared with a depth-reference floor near 10^{-15} for the projected Fréchet method.

Table 3 also records the rejected algorithmic variant. Checking the projected certificate after every candidate chooses depth 22 rather than 24, but it is 1.33–3.12 times slower than preflight because it repeatedly forms direction projections and reduced eigendecompositions. This ablation motivates the preflight residual upper bound in Theorem 3.

5. Limitations and scope. The theorem concerns real-symmetric matrices and cosine responses. Complex Hermitian matrices require conjugate transposes but no conceptual change; nonnormal generators require a numerical-range or logarithmic-norm analysis. The Taylor majorant is intentionally simple and becomes severely conservative after the actual error reaches roundoff. Spectrum-adapted polynomial or contour bounds [7, 15, 16] may reduce depth, but combining them with probe compression requires a separate proof.

The probe SVD costs $O(nb^2)$ and is useful only when b is moderate and its columns are correlated. An orthogonal or slowly decaying probe block can force $r = b$, in which case the method reduces to an ordinary full-block calculation. The Si5H12 timing demonstrates that a strict action reduction need not translate to an end-to-end CPU speedup.

Finally, the PARSEC inputs are authentic electronic-structure matrices, but the affine directions, probes, and targets are constructed numerical quantities. The study validates a matrix-function learning kernel, not an experimental Hamiltonian-identification pipeline. Device data would introduce sampling noise, model discrepancy, and likelihood design beyond the deterministic error treated here [21–23].

6. Data and code availability. The accompanying repository contains the downloader, pinned archive digests, installable CPU implementation, unit tests, complete timing arrays, locked JSON and

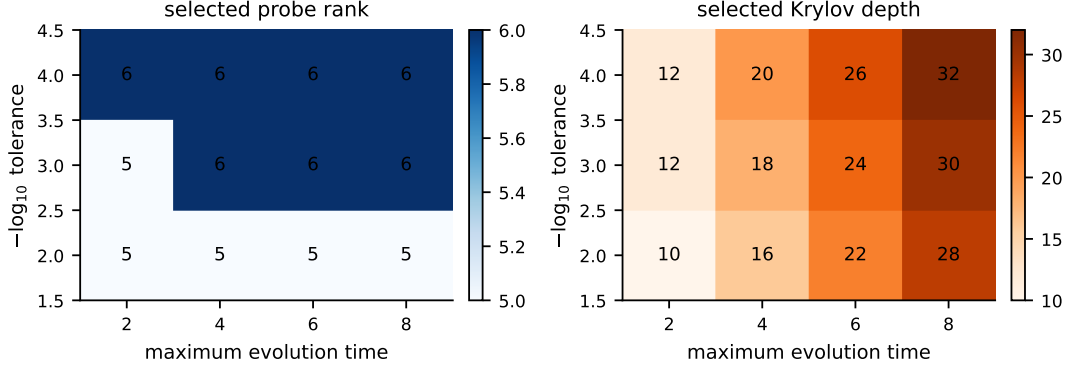


Fig. 2: Preflight choices on the Si2 development matrix over a registered time–tolerance grid. Rank reacts mainly to tolerance, whereas depth grows with both evolution time and requested accuracy. Every displayed cell satisfies the preflight certificate.

Table 2: Primary registered result at tolerance 10^{-3} . “Work” is the Hamiltonian-vector equivalent count mr . “Ratio” is the paired median time of the same-depth full block divided by the complete preflight time.

matrix	n	rank	depth	certificate	error	work	ratio
Si2	769	6	24	1.40e-04	6.87e-16	144	1.36
SiH4	5,041	5	24	2.09e-04	3.23e-15	120	1.65
benzene	8,219	6	24	1.39e-04	1.74e-15	144	2.04
Si5H12	19,896	6	24	1.47e-04	1.24e-15	144	0.99

CSV results, and figure generator. The non-timing semantic digest begins `eae62cf6af1c`. External matrices are obtained from the SuiteSparse Matrix Collection and remain subject to its terms. The repository does not redistribute the matrix archives.

7. Conclusion. Correlated quantum preparations create a block resource that should be managed before Krylov propagation, not after it. The proposed method combines an SVD tail, a uniform parameter-box enclosure, polynomial exactness, and positive cosine tails into a gradient certificate available without Hamiltonian applications. This preflight calculation selects both block rank and Krylov depth and then executes one projected gradient evaluation. The authenticated study establishes an exact work reduction and a conditional CPU benefit, while retaining the largest-matrix timing exception and the considerable slack of the certificate. These boundaries are part of the result: block compression is a certified resource option, not a universal speedup theorem.

A. Geometric evaluation of the tails. Let q be the first even integer larger than m and put $z = |t|\rho$. Consecutive terms of $R_m^{(0)}$ satisfy

$$\frac{z^{\nu+2}/(\nu+2)!}{z^\nu/\nu!} = \frac{z^2}{(\nu+1)(\nu+2)} \leq \frac{z^2}{(q+1)(q+2)} = \zeta_0. \quad (\text{A.1})$$

When $\zeta_0 < 1$,

$$R_m^{(0)}(t, \rho) \leq \frac{z^q}{q!} \frac{1}{1 - \zeta_0}. \quad (\text{A.2})$$

For derivative terms,

$$\frac{(\nu+2)|t|^{\nu+2}\rho^{\nu+1}/(\nu+2)!}{\nu|t|^\nu\rho^{\nu-1}/\nu!} = \frac{z^2}{\nu(\nu+1)} \leq \frac{z^2}{q(q+1)} = \zeta_1, \quad (\text{A.3})$$

so

$$R_m^{(1)}(t, \rho) \leq \frac{q|t|^q\rho^{q-1}}{q!} \frac{1}{1 - \zeta_1}. \quad (\text{A.4})$$

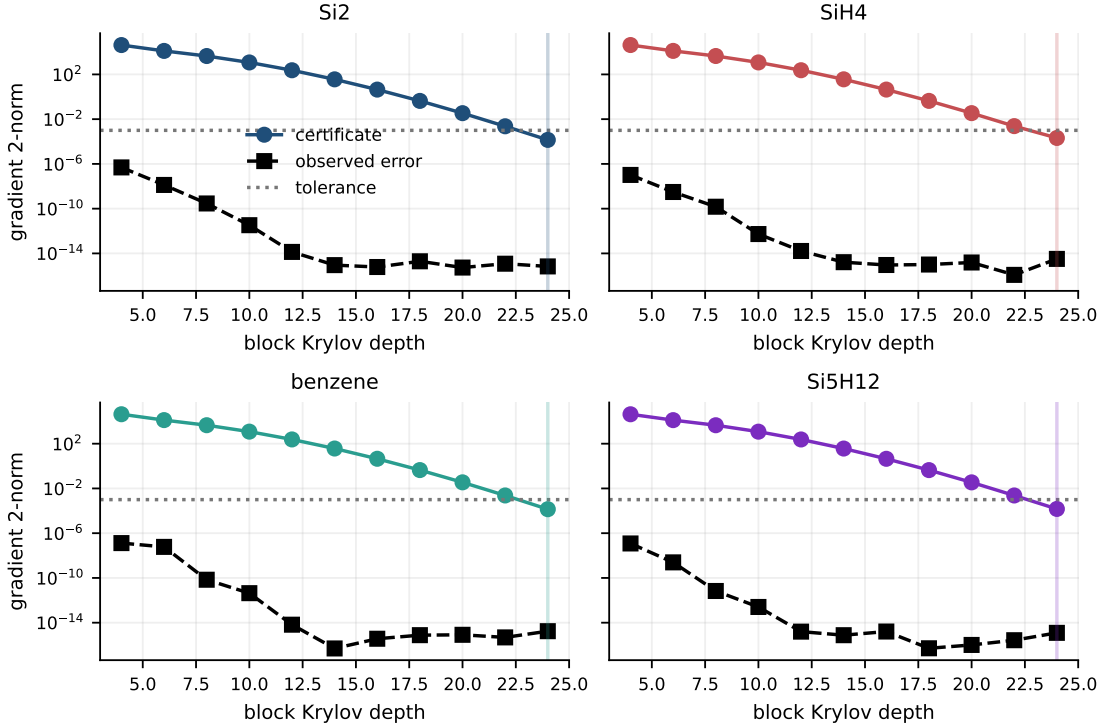


Fig. 3: Preflight certificate and observed error relative to the depth-28 full-block gradient. The dotted line is the 10^{-3} tolerance and the shaded vertical line is the first feasible depth. The certificate covers every observed error but is highly conservative after the numerical error reaches the reference floor.

Table 3: Median-time ratios relative to one preflight calculation. Values above one favor preflight; “recheck” evaluates every candidate depth.

matrix	full block	recheck	finite diff.
Si2	1.36	3.12	3.88
SiH4	1.65	1.97	7.59
benzene	2.04	1.89	10.67
Si5H12	0.99	1.33	5.28

If either ratio is at least one, that candidate depth is declared infeasible and the preflight scan advances. The implementation uses 80-digit decimal arithmetic for these expressions and rounds the binary64 conversion toward $+\infty$.

B. Reference and finite-difference checks. The depth reference is not used by the stopping rule. It appears only in the post hoc error plots and tables. For each matrix, targets are recomputed at depths 26 and 28 with the full eight-column block. The same comparison is made for the analytical projected gradient. Central differences use $h = 10^{-5}$ and rebuild the loss at $\theta_0 \pm h e_j$; they do not reuse a basis or derivative projection. Thus their *2pmb* action ledger is an executed baseline rather than an extrapolation.

References.

- [1] N. J. Higham, *Functions of Matrices: Theory and Computation* (Philadelphia, 2008), doi:10.1137/1.9780898717778.
- [2] Y. Saad, Analysis of some Krylov subspace approximations to the matrix exponential operator, vol. **29**, pp. 209–228 (1992), doi:10.1137/0729014.
- [3] M. Hochbruck and C. Lubich, On Krylov subspace approximations to the matrix exponential operator, vol. **34**, pp. 1911–1925 (1997), doi:10.1137/S0036142995280572.

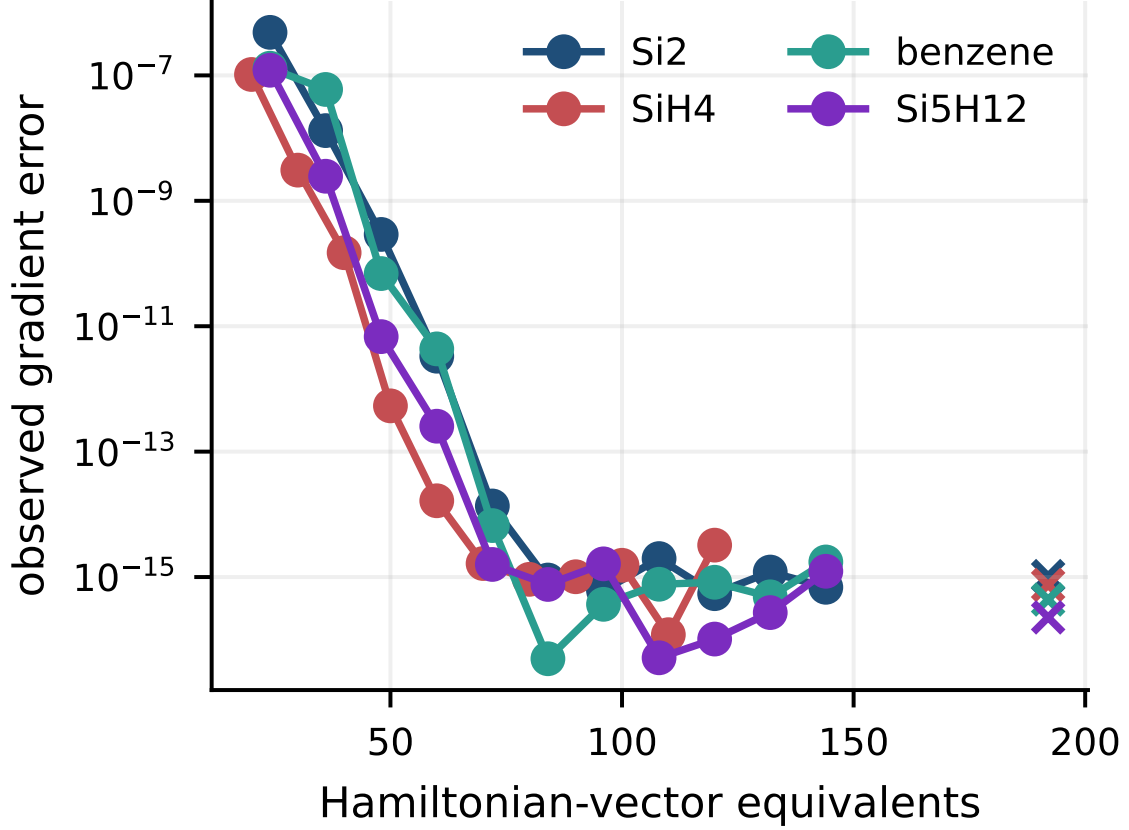


Fig. 4: Observed gradient error versus Hamiltonian-vector equivalents along the compressed depth frontier. Crosses are full-block evaluations at the selected depth. Compression reduces deterministic work while both methods reach the reference floor.

- [4] C. Lanczos, An iteration method for the solution of the eigenvalue problem of linear differential and integral operators, *J. Res. Natl. Bur. Stand.* **45**, 255 (1950).
- [5] G. H. Golub and G. Meurant, *Matrices, Moments and Quadrature with Applications* (Princeton University Press, Princeton, 2010).
- [6] M. H. Gutknecht, Block Krylov space methods for linear systems with multiple right-hand sides: an introduction, in *Modern Mathematical Models, Methods and Algorithms for Real World Systems*, edited by A. H. Siddiqi, I. S. Duff, and O. Christensen (Anamaya, New Delhi, 2007), pp. 420–447.
- [7] Q. Xu and T. Chen, A posteriori error bounds for the block-Lanczos method for matrix function approximation, *arXiv:2211.15643* (2022).
- [8] P. Kandolf and S. D. Relton, A block Krylov method to compute the action of the Fréchet derivative of a matrix function on a vector with applications to condition number estimation, vol. **39**, pp. A1416–A1434 (2017), doi:10.1137/16M1077969.
- [9] P. Kandolf, A. Koskela, S. D. Relton, and M. Schweitzer, Computing low-rank approximations of the Fréchet derivative of a matrix function using Krylov subspace methods, *Numer. Linear Algebra Appl.* **28**, e2401 (2021).
- [10] D. Kressner, A Krylov subspace method for the approximation of bivariate matrix functions, in *Structured Matrices in Numerical Linear Algebra* (Springer, Cham, 2019), pp. 197–214.
- [11] N. Singh, K. Barros, and X. S. Li, Fast and stable gradient approximation for bilinear forms of Hermitian matrix functions, *arXiv:2605.12801* (2026).

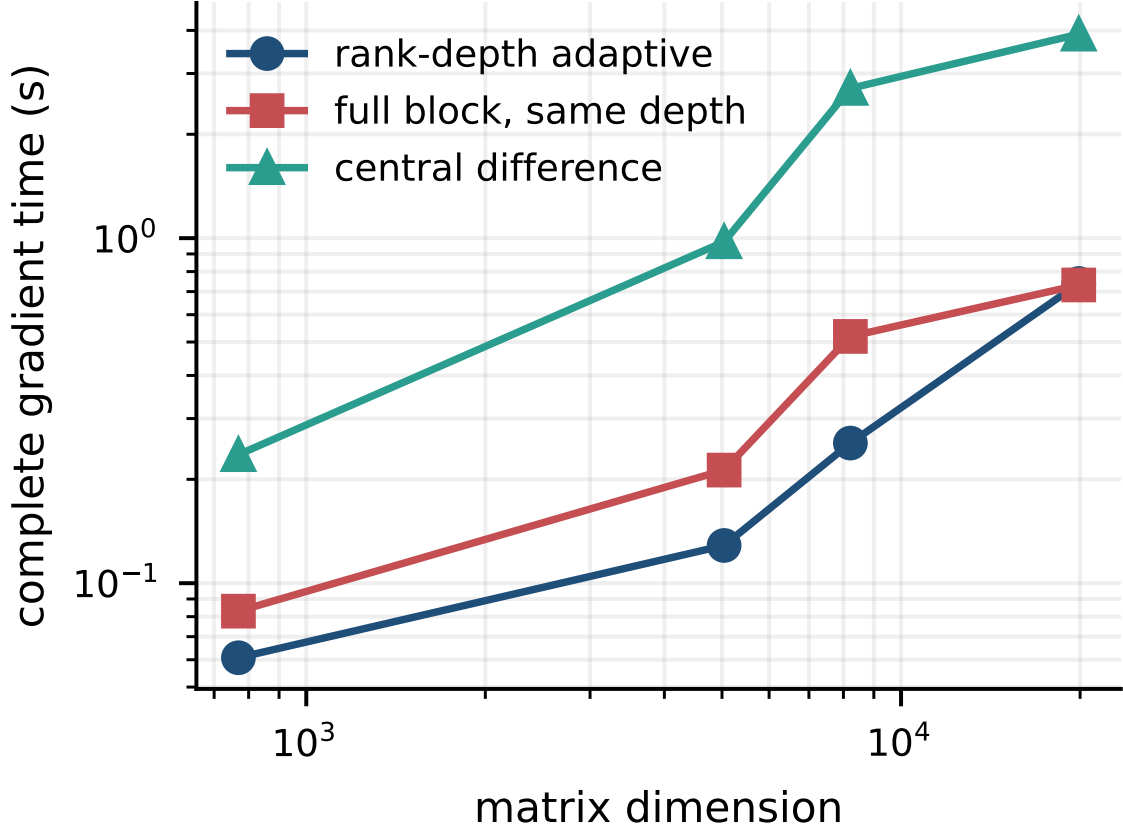


Fig. 5: Complete gradient time against matrix dimension. Lines connect the four distinct matrices only as visual guides. On Si5H12 the full block is 1 percent faster than the proposed method despite requiring more Hamiltonian-vector equivalents.

- [12] A. H. Al-Mohy and N. J. Higham, Computing the Fréchet derivative of the matrix exponential, with an application to condition number estimation, vol. **30**, pp. 1639–1657 (2009), doi:10.1137/080716426.
- [13] T. A. Davis and Y. Hu, The University of Florida Sparse Matrix Collection, ACM Trans. Math. Softw. **38**, 1 (2011).
- [14] J. R. Chelikowsky, N. Troullier, and Y. Saad, Finite-difference-pseudopotential method: Electronic structure calculations without a basis, Phys. Rev. Lett. **72**, 1240 (1994).
- [15] T. Chen, A. Greenbaum, C. Musco, and C. Musco, Error bounds for Lanczos-based matrix function approximation, vol. **43**, pp. 787–811 (2022), doi:10.1137/21M1427784.
- [16] I. Simunec, Error bounds for the approximation of matrix functions with rational Krylov methods, arXiv:2311.02701 (2023).
- [17] B. Beckermann, D. Kressner, and M. Schweitzer, Low-rank updates of matrix functions, vol. **39**, pp. 539–565 (2018), doi:10.1137/17M1140108.
- [18] N. J. Higham and S. D. Relton, Higher order Fréchet derivatives of matrix functions and the level-2 condition number, vol. **35**, pp. 1019–1037 (2014), doi:10.1137/130945259.
- [19] E. H. Rubensson, A unifying framework for higher order derivatives of matrix functions, vol. **45**, pp. 504–528 (2024), doi:10.1137/23M1580589.
- [20] C. F. Van Loan, Computing integrals involving the matrix exponential, IEEE Trans. Autom. Control **23**, 395 (1978).

- [21] A. Shabani, R. L. Kosut, M. Mohseni, H. Rabitz, M. A. Broome, M. P. Almeida, A. Fedrizzi, and A. G. White, Efficient measurement of quantum dynamics via compressive sensing, *Phys. Rev. Lett.* **106**, 100401 (2011).
- [22] N. Wiebe, C. Granade, C. Ferrie, and D. G. Cory, Hamiltonian learning and certification using quantum resources, *Phys. Rev. Lett.* **112**, 190501 (2014).
- [23] E. Bairey, I. Arad, and N. H. Lindner, Learning a local Hamiltonian from local measurements, *Phys. Rev. Lett.* **122**, 020504 (2019).
- [24] M. Cramer, M. B. Plenio, S. T. Flammia, R. Somma, D. Gross, S. D. Bartlett, O. Landon-Cardinal, D. Poulin, and Y.-K. Liu, Efficient quantum state tomography, *Nat. Commun.* **1**, 149 (2010).
- [25] A. M. Childs, D. Maslov, Y. Nam, N. J. Ross, and Y. Su, Toward the first quantum simulation with quantum speedup, *Proc. Natl. Acad. Sci. USA* **115**, 9456 (2018).